

Confidently Wrong: Handing a Coding Agent the Database Anyway

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A year of daily notes is 220 Markdown files

```
● ● ● ~/notes
$ ls notes/
2026-07-08.md 2026-07-23.md 2026-08-07.md 2026-08-24.md 2026-09-09.md 2026-09-28.md
2026-07-09.md 2026-07-24.md 2026-08-10.md 2026-08-25.md 2026-09-11.md 2026-09-30.md
2026-07-10.md 2026-07-27.md 2026-08-11.md 2026-08-26.md 2026-09-14.md 2026-10-01.md
2026-07-13.md 2026-07-28.md 2026-08-12.md 2026-08-27.md 2026-09-15.md 2026-10-02.md
2026-07-14.md 2026-07-29.md 2026-08-13.md 2026-08-31.md 2026-09-16.md 2026-10-05.md
2026-07-15.md 2026-07-30.md 2026-08-14.md 2026-09-01.md 2026-09-17.md 2026-10-06.md
2026-07-16.md 2026-07-31.md 2026-08-17.md 2026-09-02.md 2026-09-18.md 2026-10-07.md
2026-07-17.md 2026-08-03.md 2026-08-18.md 2026-09-03.md 2026-09-21.md archive
2026-07-20.md 2026-08-04.md 2026-08-19.md 2026-09-04.md 2026-09-23.md
2026-07-21.md 2026-08-05.md 2026-08-20.md 2026-09-07.md 2026-09-24.md
2026-07-22.md 2026-08-06.md 2026-08-21.md 2026-09-08.md 2026-09-25.md
$ find notes -name "*.md" | wc -l
220
```

rg can't find a note I only half remember

```
~ /notes
$ rg -l "migration without a copy of production" notes/
$ rg -l migration notes/ | wc -l
20
```

I can ask my coding agent to look. It works, and it spends tokens I'd rather spend on code.



**Why not have a
coding agent build
the fix?**

Good ideas are always crazy
until they're not.

Larry Page

I wrote one page about what I want



talk/notes-app-spec.md

Why

I keep my notes as Markdown files in a folder, one file per day, named with a date stamp. After a year there are hundreds of files. So, each quarter, I summarize the work and move that quarter's files into an archive folder.

Finding things is a problem. `rg` and `fzf` find exact words, but I rarely remember the words. **I remember that a note was good**, that it has the facts I need, and roughly what it was about. Asking a coding agent to dig through the files works, but it burns tokens I would rather spend on code.

Stack

- MariaDB 12.3

- Everything runs on my laptop with no API keys. **I don't want my notes** to leave my laptop.

The spec lists what to build now and what can wait

```
talk/notes-app-spec.md

## Must have

1. Import every note, current and archived
2. Pick up new and changed notes when I run the import again
3. Search by meaning, and show the best matches with their title, date,
   and folder
4. Search for exact words too
5. Open a note and render its Markdown
6. Open the note in `nvim` if I hit enter
7. Search from the command line
8. Keybindings like `lazygit`, and the bottom bar is always visible

## Nice to have

1. `cron` automation
2. MCP server access
3. Standardized logging set through debug levels
```

The spec ends with a check the app must pass

```
● ● ● talk/notes-app-spec.md
## Done when

- Every note file is in the database
- Searching for "testing a migration without a copy of production" shows my
April 14, 2026 note in the top three
```

The April 14 note never uses those words. If the app finds it, the app searches by meaning.

First try: Claude Haiku 4.5 designs the database



- 1. Design the MariaDB schema and save it as working/notes_app.sql
- 2. Write the SQL to search my notes by meaning and by exact words
- 3. Tell me how safe the schema is to run more than once

Its search ran and put the exact match last

```

MariaDB 12.3.3 sandbox
MariaDB [haiku_n]> -- Claude Haiku 4.5's search, three test notes,
MariaDB [haiku_n]> -- query vector identical to closest.md
+-----+-----+
| title      | distance_score |
+-----+-----+
| far.md     | 0.000000       |
| middle.md  | 0.000000       |
| closest.md | 1.000000     |
+-----+-----+
```

The search used another database's distance operator. In MariaDB, that operator asks whether two values are identical.

The agent called reruns safe, but they skip changes

Claude Haiku 4.5, then MariaDB 12.3.3

- Safety of running the DDL multiple times: ✓ **Fully safe and idempotent.**

```
MariaDB [haiku_ns]> CREATE TABLE IF NOT EXISTS notes ( ... note_date DATE NULL ... );  
MariaDB [haiku_ns]> SHOW WARNINGS;
```

Level	Code	Message
Note	1050	Table 'notes' already exists

```
MariaDB [haiku_ns]> SELECT COLUMN_NAME, IS_NULLABLE FROM information_schema.COLUMNS ...
```

COLUMN_NAME	IS_NULLABLE
note_date	NO

MariaDB's plugin adds knowledge and a live server



Skills: what the agent knows

- MariaDB's current syntax, in the agent's context
- The order statements have to run in
- The mistakes the docs don't warn you about

MCP server and sandbox: where it checks

- Runs SQL and reads the schema on a live server
- Deploys a MariaDB sandbox in one tool call
- No Docker and no root

Two commands to install, one to give it access

```
/plugin marketplace add mariadb/ai-plugins  
/plugin install dev@mariadb
```

```
$ mariadb-shell -- mcp setup --addPaths=~/.notes-app
```

Folders the MCP server may read and write

```
$ mariadb-shell -- mcp setup --addConnection=demo@localhost:3306
```

Database servers it may connect to, with the password kept by setup

Both lists start empty. The agent adds only the sandboxes it deploys.

With the skills loaded, the exact match comes first

```

MariaDB 12.3.3 sandbox
MariaDB [haiku_n]> -- the search form from the skills-loaded run,
MariaDB [haiku_n]> -- same three notes, same query vector
+-----+-----+
| title   | distance |
+-----+-----+
| closest.md | 0.000 |
| middle.md | 0.293 |
| far.md   | 1.000 |
+-----+-----+

```

Same agent, same prompt. Asked about reruns, it now says:
“Second run wipes the database.”

Check that the agent actually loaded the skill

```
● ● ● Claude Code: Haiku 4.5  
> /dev:mariadb-vector  
● Skill(dev:mariadb-vector)  
  └─ Successfully loaded skill
```

Haiku 4.5 didn't load the skills on its own, even when my prompt asked it to. I loaded them myself.

Skill lines carry knowledge. Tool calls act on the server.

For the full build, I switched to Claude Opus 5.5



Claude Code: Opus 5.5

- > Please run these steps
Work in this repository and **complete every step in order**. Write your files to working/, and append a short record of each step to working/RUN_LOG.md.
1. Design the MariaDB schema for the app that talk/notes-app-spec.md describes, and **save the DDL as working/notes_app.sql**.
 2. If no MariaDB 12.3 sandbox is running on port 3310, deploy one there with root password demo-pw and data directory working/sandbox. Run working/notes_app.sql on it via the MCP server.
 3. Report each table you created and what it holds.

Name the file. Number the steps. Say “in order”.

Next in the transcript: four MariaDB Skill lines, before the agent wrote anything.

The agent built the app in about seven minutes

Two prompts, Claude Opus 5.5 with the MariaDB plugin

- Designed the schema and deployed a sandbox
- Imported all 220 notes, 351 sections embedded
- Built the terminal app and the search command

Time per prompt

2:12

Schema and sandbox, plus one question for me

5:07

App, import, and the "Done when" check

The agent tested its own work and fixed what it found



Claude Code: Opus 5.5

- The editable install **ended up empty** since uv built the wheel with its own backend instead of respecting hatchling's packages setting. I'll switch to uv_build and name the module explicitly to fix this.
...
- The TUI test passes fully, but it **revealed a bug in exact-word mode**: required stopwords like "a" and "of" aren't stored in InnoDB's full-text index, so queries containing them return 0 results. I'll fix to_boolean_query to exclude those words.

My check passes

```
●●● ~/notes-app
$ uv run notes-app search "testing a migration without a copy of production" -n 3
1. 0.633 2026-04-14 Tuesday, April 14, 2026
      notes/archive/2026-Q2/2026-04-14.md > Hallway track
2. 0.628 summary      Q2 2026 Quarterly Summary
      notes/archive/2026-Q2/2Q2026.md > APRIL 2026
3. 0.623 2025-10-23 Thursday, October 23, 2025
      notes/archive/2025-Q4/2025-10-23.md > Migrator demo notes
```

April 14 ranks 1st, just ahead of the Q2 summary that describes the same conversation.

Now I can search for what I remember

notes-app

Meaning: "rehearsing a database move with fake data": 15 notes | mode: meaning (m to switch)
rehearsing a database move with fake data

Date	Title
2026-04-14	Tuesday, April 14, 2026
2026-07-15	Wednesday, July 15, 2026
2025-10-08	Wednesday, October 8, 2025
2026-05-20	Wednesday, May 20, 2026
2026-05-06	Wednesday, May 6, 2026
summary	Q2 2026 Quarterly Summary
2025-12-09	Tuesday, December 9, 2025
2026-02-26	Thursday, February 26, 2026
2025-10-23	Thursday, October 23, 2025
2025-10-02	Thursday, October 2, 2025
2026-02-11	Wednesday, February 11, 2026
2026-09-02	Wednesday, September 2, 2026
summary	Q4 2025 Quarterly Summary
summary	Q1 2026 Quarterly Summary
2026-03-06	Friday, March 6, 2026

Tuesday, April 14, 2026

Stuff

- [x] Badge pickup before 8
- [x] Session at 11:15 -- went long, cut the second demo
- [] Send slides to the organizers
- [] Expense the cab from the airport

Hallway track

Tags: #talk-idea #migration

Three different people came up after the session with the same problem: how do you rehearse a database move

Open in nvim q Quit / Search m Mode ? Help palette

Add one line in CLAUDE.md to use my new tool

```
~ /notes-app
$ cat CLAUDE.md
# Notes

To find one of my notes, ask the index first, with my words as the query:
`uv run notes-app search "<my words>". Read a note file only after the
search names it.
```

Without the line: grep and read, \$0.43

With the line: ask the index first, \$0.25

Same app, same question, one run each

The harness stopped a delete on its own

```
DELETE FROM notes_app.account
```

No WHERE clause. The foreign keys cascade to every other table.

“Permission for this action was denied by the Claude Code auto mode classifier. Reason: [Cloud Storage Mass Delete].”

Three layers decide what the agent can run

The allow-list

- Which folders and database servers the agent can reach
- Set by you, with mcp setup

The harness

- Which actions the agent can take
- Stopped this delete

The database grants

- What the account may do
- Would have allowed it

Five things make codeveloping with an agent easier

- Adopt spec driven development (SDD)
 - Write down what the app does, and the check it must pass
- Name the files, number the steps, and tell the harness to do things "in order"
- Load skills and plugins for your stack
 - Look for the Skill line to make sure the harness is paying attention!
- Give the agent a sandbox to try its work on
 - Always work in isolation when rapid-prototyping
 - Create snapshot clones for experimentation
- Run the check yourself and provide feedback to the agent



**Go build the tool you've been
missing, on open source you own**



Demo: from the spec to a working search



github.com/maglietti/notes-app-demo
ai-plugins.mariadb.org